

Probabilistic Graphical Model

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1 Introduction

Probabilistic graphical models expresses probability distribution in a graph. This provides us several useful properties:

- Provide a simple way to visualize the structure of a probabilistic model.
- Properties of the model, such as conditional independence, can be obtained by inspecting the graph.
- Complex computations such as inference and learning can be expressed in terms of manipulating graphs.

The graph consists of nodes and links (or vertices and edges). In probabilistic graphical models, each node represents a random variable and the links express probabilistic relationship between nodes. There are two types of probabilistic graphical models which differ in the way of representation of the relationship between nodes. The **Bayesian network** uses directed acyclic graph, and the **Markov random field** uses undirected graph for links.

2 Bayesian Network

2.1 Definition

Definition 2.1.1 (Bayesian Network)

Let $G = (V, E)$ be an acyclic, directed graph, where $V = \{\mathbf{x}_1, \dots, \mathbf{x}_K\}$. Let pa_k be the set of **parent** nodes of $\mathbf{x}_k$. Then, the **Bayesian network** with graph G represents the following probability distribution;

$$p(\mathbf{x}) = \prod_{k=1}^K p(\mathbf{x}_k | pa_k)$$

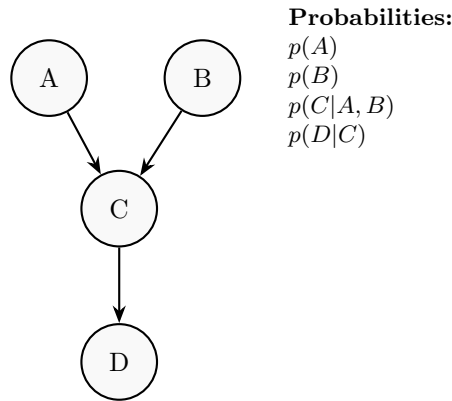


Figure 1: Bayesian network

Figure 1 represents the probability distribution $p(A, B, C, D) = p(D|C)p(C|A, B)p(B)p(A)$. We will discuss about graphical notations in next section.

2.2 Graphical Notations

TODO

2.3 Conditional Independences

2.3.1 D-seperation

Before introducing the D-seperation method, which is useful for examining conditional independence properties of given bayesian network, we define some terminologies;

- head-to-head node with respect to path a : The node connects to the heads of the two arrows of the link in the path.
- head-to-tail node with respect to path a : The node connects to the head and tail of two arrows of the link in the path.
- tail-to-tail node with respect to path a : The node connects to the tails of the two arrows of the link in the path.
- descendant of node $\mathbf{x}$: There is a path to the node starting from $\mathbf{x}$, considering the directions of the link.

Theorem 2.3.1 (D-seperation)

Consider a set of nodes A , B , and C , which does not intersect each other. Then, the path (ignoring the directions of links) from some element of A to some element of B is **blocked** by C if has node that satisfies one of following two conditions:

1. head-to-tail or tail-to-tail with respect to the path and belongs to C ;
2. head-to-head with respect to the path or the descendant of such node and does not belong to C .

If every path is blocked, then the following conditional independence is satisfied:

$$A \perp\!\!\!\perp B \mid C$$

2.3.2 Markov Blanket

The **Markov blanket** is defined to be the nodes that make a given node to be conditionally independant to every other nodes when observed, i.e., the set of nodes M_i that satisfy

$$p(x_i, x_j | M_i) = p(x_i | M_i) p(x_j | M_i)$$

for every j . In Bayesian network, the Markov blanket of node x_i consists of:

- parents;
- children;
- and co-parent of children.

3 Markov Random Field

3.1 Definition

Before defining Markov random field, we define some concepts.

Definition 3.1.1 (Clique)

Clique is a subset of nodes which have links connected to all the other nodes in the subset. **Maximal clique** is a clique which cannot add any more nodes without being clique anymore.

Now we define Markov random field.

Definition 3.1.2 (Markov Random Field)

Let $G = (V, E)$ be an undirected graph. Then, the **Markov random field** with graph G represents the following probability distribution:

$$p(\mathbf{x}) = \frac{1}{Z} \prod_C \psi_C(\mathbf{x}_C)$$

where C ranges over all maximal clique of G and $\mathbf{x}_C$ is all the node in C . *partition function*, Z , is the constant that normalizes the equation so that it can represent a probability distribution. ψ_C is called *potential function*, and can be any function of $\mathbf{x}_C$ with condition that it is positive or strictly positive.

3.2 Conditional Independence

3.2.1 Conditional Independence in Markov Random Field

The conditional independence is easier to examine in Markov random field.

Theorem 3.2.1 (Conditional Independence in Markov Random Field)

Let A , B , and C be a set of nodes in Markov random field. Consider a path from A to B . Then, the path is **blocked** by C if the path contains one or more nodes in C . If all path is blocked by C , then $A \perp\!\!\!\perp B \mid C$.

Simple way is to think that $A \perp\!\!\!\perp B \mid C$ when nodes in A cannot reach any node in B if we delete all the nodes of C .

3.2.2 Markov Blanket

The Markov blanket of a given node in MRF is simply the adjacent nodes of a given node.

4 $\mathcal{I}$ -map, $\mathcal{D}$ -map and $\mathcal{P}$ -map

We now know how to represent probability distribution in terms of graphs, and we can do the opposite task, too. However, not all the probabilistic informations are preserved when converting from one to other. We propose a concept which describes this situation.

4.1 $\mathcal{I}$ -map

Definition 4.1.1 ($\mathcal{I}$ -map)

Let $p(\mathbf{x})$ be a probability distribution. Then, the $\mathcal{I}$ -map of the distribution is a graph whose conditional independencies are all satisfied by the distribution.

That is, $\{\text{probabilistic information of } \mathcal{I}\text{-map} \subset \text{probabilistic information of } p(\mathbf{x})\}$.

4.2 $\mathcal{D}$ -map

Definition 4.2.1 ($\mathcal{D}$ -map)

Let $p(\mathbf{x})$ be a probability distribution. Then, the $\mathcal{D}$ -map of the distribution is a graph where all the conditional independencies in the distribution can be recognized from the graph itself.

That is, $\{\text{probabilistic information of } p(\mathbf{x}) \subset \text{probabilistic information of } \mathcal{D}\text{-map}\}$.

4.3 $\mathcal{P}$ -map

Definition 4.3.1 ($\mathcal{P}$ -map)

Let $p(\mathbf{x})$ be a probability distribution. Then, the $\mathcal{P}$ -map of the distribution is a graph which is simultaneously $\mathcal{I}$ -map and $\mathcal{D}$ -map of the distribution.

That is, $\{\text{probabilistic information of } \mathcal{P}\text{-map} = \text{probabilistic information of } p(\mathbf{x})\}$.

5 Inferences in Probabilistic Graphical Models

[See this document.](#)

6 Application

6.1 Linear-Gaussian Models

Definition 6.1.1 (Linear-Gaussian Model)

The **linear-Gaussian model** is a Bayesian network with following properties;

- Each node follows a Gaussian distribution.
- The mean of the Gaussian distribution is a linear combination of the parent nodes, i.e., $p(\mathbf{x}_i | pa_i) = \mathcal{N}\left(\mathbf{x}_i \mid \sum_{j \in pa_i} \mathbf{W}_{ij} \mathbf{x}_j + \mathbf{b}_i, \Sigma_i\right)$

Due to the properties of Normal distribution, the whole probability distribution $p(\mathbf{x})$ is again a Normal distribution.

The conjugate prior is a special case of the linear-Gaussian model.

6.2 Hidden Markov Model

6.3 Linear Dynamic System